

YaadSaathi: Definitive Userbase, Clinical Dementia & Therapy Solutions, and Advanced Feature Architecture

AI-Powered Cognitive Gaming, Reminiscence Therapy, and Multi-Stakeholder Neuro-Rehabilitation Platform

Target Platform

YaadSaathi (Wolf-Bazil/Yaad-Saathi)

Core Deployment Region

North East Region & Rural PHCs, India

Specialist Disparity

<1 Geriatric Neurologist / 250k pop

Feature Architecture

/teamwork-preview & /boost

1 Executive Summary & Core Clinical Mandate

YaadSaathi (यादसाथी) is an indigenous, offline-first digital therapeutics and cognitive rehabilitation platform. It bridges a profound healthcare disparity in rural and semi-urban India—specifically the North East Region (NER)—where geographical isolation, over 200 dialects, terrain power/fiber cuts, and an acute specialist deficit leave millions of neurological patients without institutional care.

The Fundamental Dual-Cohort Paradigm:

- **Dementia Care is Decelerative & Compensatory:** Irreversible neurodegeneration requires synaptic reserve preservation, errorless learning, agitation/sundowning reduction, and acute emergency detection (UTI/sepsis/stroke).
- **Active Therapy is Restorative & Neuroplastic:** Focuses on circuit re-engagement, massed practice compliance, lexical retrieval in aphasia, motor pacing, and quantifiable functional recovery over 6–24 week cycles.

2 The Multi-Tiered Userbase Ecosystem

Primary Cohort A

Dementia & Neurodegenerative Patients

- **Amnesic Mild Cognitive Impairment (MCI):** Preserved ADLs, delayed recall erosion.
- **Early-to-Moderate Alzheimer's (AD):** Prosopagnosia (family face amnesia), word-finding gaps.
- **Vascular Dementia (VaD):** Multi-infarct micro-strokes, patchy executive slowing.
- **Lewy Body & Parkinson's Dementia:** Day/night lucidity fluctuations, tremors, visual vulnerability.
- **Frontotemporal Dementia (FTD):** Executive sequencing loss, speech preservation needs.
- **Adults with Down Syndrome:** Early-onset late-life Alzheimer's neuropathology.

Primary Cohort B

People in Active Neuro-Rehabilitation Therapy

- **Post-Stroke Rehabilitation:** Middle Cerebral Artery (MCA) survivors, spatial neglect.
- **Speech-Language Therapy (Aphasia):** Broca's & Anomic aphasia with lexical blocks.
- **Traumatic Brain Injury (TBI):** Post-concussive fatigue, divided attention collapse.
- **Neuro-Occupational Therapy (OT):** ADL retraining (chai, medication, hygiene).
- **Geriatric Psychotherapy:** Reminiscence therapy for late-life depression & pseudo-dementia.
- **Chemobrain (CRCI):** Oncology survivors regaining processing speed.

Secondary Tier

Informal Family Caregivers

Adult daughters, spouses, and grandchildren bearing **85%+ of dementia care unpaid**. Need remote monitoring and burnout relief.

Tertiary Tier

Grassroots Cadre (ASHA / ANM)

Frontline staff at Ayushman Arogya Mandir (Sub-Centres) with sub-₹8,000 Android phones, zero specialists within 180km.

Quaternary Tier

Clinicians & Prescribers

SLPs, OTs, Psychiatrists, and Medical Officers requiring objective sub-millisecond telemetry to verify between-session home adherence.

3 Clinical Paradigm Comparison Matrix

Dimension	Dementia Patient Cohort	People in Active Therapy
Primary Clinical Goal	Deceleration of cognitive decay, dignity preservation	Restoration of damaged neural pathways, neuroplastic rewiring
Neurological Target	Reinforcing remote autobiographical engrams	Re-engaging disrupted synaptic circuits (aphasia, motor, OT)
Learning Modality	100% Errorless Learning (zero failure buzzers)	Calibrated challenge pacing & progressive cognitive resistance
Engagement Horizon	Indefinite daily maintenance therapy	6 to 24 week structured recovery cycles
Primary Hazard	Catastrophic panic reactions, sundowning delirium	Between-session non-compliance, neuro-fatigue regression
Core Telemetry Metric	Latency variance, anomaly detection (UTI/sepsis alerts)	Reaction speed gain, 5-spoke cognitive radar elevation

4 What YaadSaathi Solves Exactly in Dementia Patients

Dementia Challenge	Traditional Deficit / App Failures	What YaadSaathi Solves Exactly	Code Engine
Prosopagnosia & Family Amnesia	Western brain apps use abstract shapes and clip-art, confusing rural elders.	Autobiographical Memory Vault: Re-activates remote cortical engrams using authentic caregiver photos with positive cues.	ClinicalPillars.jsx Pillar 2 (Vault)
"Catastrophic Reaction" & Panic	Loud error buzzers (BUZZ!) and countdown clocks trigger tears, panic, and app abandonment.	Zero-Buzzer Errorless Learning: No negative tones exist. Adaptive engine drops difficulty before frustration occurs.	audio.js#L59 RandomForest ML
Visuospatial Disorientation	Matching foreign chess sets or London clocks causes emotional alienation.	Cultural Memory Match: Parahippocampal gyrus stimulation with regional motifs (Jorhat Kettle, Bihiu Japi, Rhino).	ClinicalPillars.jsx Pillar 1 (Culture)
Executive Sequencing Decay	Complex multi-level menus overload depleted working memory buffers.	Sequence Pattern Memory: Progressively calibrated flash cues stimulating the dorsolateral prefrontal cortex (DLPFC).	ClinicalPillars.jsx Pillar 3 (Sequence)
Loss of ADL Independence	Games don't translate to real-world survival habits.	Daily Routine Association: Circadian temporal markers (08:00 AM) reinforce vital habits (chai, morning meds).	ClinicalPillars.jsx Pillar 4 (ADLs)
Circadian Sundowning Delirium	Notifications pop up during late-afternoon agitation windows.	Sundowning Clusterer (K-Means): Identifies peak lucidity window (10:00–11:30 AM) and locks gameplay to that period.	CapabilityGrid.jsx Model 3 (K-Means)
Hidden Acute Emergencies	Sudden apathy is chalked up to dementia, missing fatal UTIs, sepsis, or micro-strokes.	Psychomotor Anomaly Detector: Detects >300% tap latency spikes and fires emergency Flag -1 alert to local ASHA.	AiArchitecture IsolationForest
Mountainous Network Cuts	Cloud apps freeze with loading spinners during frequent rural fiber cuts.	100% Offline IndexedDB Engine: Entire gameplay, voice TTS, and logging run locally; auto-flushes in 280ms on reconnect.	OfflineEngine IndexedDB + Sync

5 What YaadSaathi Solves Exactly in People in Active Therapy

1. Bridging the "Between-Session Compliance Chasm"

Clinical Need: Neuroplastic reorganization requires hundreds of trials per week. Clinical sessions occur only 1–2x weekly for 30 mins. **Over 70% of patients abandon paper exercise handouts** due to boredom and lack of feedback.

YaadSaathi Solution: Gamified 10-minute micro-sessions with harmonic pentatonic chord feedback (audio.js) and canvas confetti provide self-directed dopamine rewards, driving 80%+ weekly home compliance.

2. Objective Millisecond Telemetry vs. Subjective Recall

Clinical Need: Therapists must rely on inaccurate family recall ("*He seemed a bit better on Thursday*") to adjust clinical protocols.

YaadSaathi Solution: Delivers raw, scientific clinical telemetry: 30-day tap latency distribution curves, standard deviation metrics, and a 5-spoke cognitive radar (Memory, Attention, Recognition, Speed, Reasoning) with exportable ICMR-compliant bilingual PDF reports.

3. Tri-Channel Lexical Retrieval for Aphasia / SLT

Clinical Need: Post-stroke anomic aphasia patients know the concept they want to articulate, but the neural bridge to the word-form is blocked.

YaadSaathi Solution: 3-tier sensory cueing hierarchy: (1) Authentic visual photograph, (2) Spoken audio prompt at 0.88x speed cadence via Web Speech API, and (3) Errorless multiple-choice selection cards providing phonological and semantic scaffolding.

4. Dynamic Fatigue Thresholding in Neuro-OT

Clinical Need: TBI and stroke patients suffer from sudden cognitive exhaustion. Pushing through fatigue triggers motor tremors and regression.

YaadSaathi Solution: Continuous latency monitoring: when tap latency exceeds 2.5s, difficulty automatically downshifts to *Level 2 (Calibrated Relief)*, protecting neural integrity while avoiding patient distress.

5. Autobiographical Reminiscence for Late-Life Depression & Pseudo-Dementia

Severe late-life depression and grief cause profound cognitive apathy, frequently misdiagnosed as permanent Alzheimer's. YaadSaathi's reminiscence exercises anchor patients in positive life memories (ancestral homes, weddings, festivals), validating identity, activating mesolimbic dopamine pathways, and restoring self-worth.

6 Feature Deep-Dive: /teamwork-preview

Clinical Problem Solved: Generative AI models synthesize quizzes directly from raw caregiver photos. In vulnerable patients, an unvetted question featuring a deceased relative or an overly complex linguistic prompt can trigger acute bereavement trauma, wandering, or catastrophic panic.

Mode 1

Care-Team Staging Sandbox

A pre-flight verification dashboard connecting Caregiver, SLP/OT, and ASHA worker:

- **Caregiver Emotional Veto:** 1-click clearance ([✓ Safe] vs [⚠ Sensitive]).
- **Therapist Difficulty Dial:** Option density (2 choices for aphasia vs 4 standard) and exposure limits.
- **ASHA Dialect Audition:** Auditions Assamese/Hindi audio pronunciation.

Mode 2

Intergenerational Co-Op Duo

A dual-view collaborative interface for two participants (Grandchild + Grandparent):

- **Companion View:** Shows conversational prompts ("Ask Dadi about the brass kettle").
- **Elder View:** High-contrast 72px tactile answer tiles.
- **Shared Harmony:** Multi-speaker pentatonic chord celebration upon mutual completion.

Mode 3

Remote Tele-Rehab Shadowing

Enables a clinical specialist in Guwahati to shadow a patient in a remote village Sub-Centre:

- Sub-200ms telemetry synchronization via Supabase Realtime data channels.
- Therapist live latency monitor and remote session pause during signs of distress.
- Remote "Gentle Pointer" to highlight cards for guided learning.

7 Feature Deep-Dive: /boost

Clinical Problem Solved: Elevates the static visual simulation of `cstBoosted` in `TelemetryPreview.jsx` into an active, evidence-backed Cognitive Stimulation Therapy (CST) and neuroplasticity accelerator engine.

Engine 1

Circadian 3-Minute Micro-Boost

Replaces exhausting 30-minute cognitive drills with focused 3-minute warmups scheduled strictly during the elder's peak lucidity window (10:00–11:30 AM). Primes prefrontal acetylcholine pathways before medication or physical therapy.

Engine 2

Adaptive Scaffolding Lifeline

Auto-intervenes when patient hesitation exceeds 2.5s: dims 2 incorrect distractors, triggers a pre-recorded caregiver voice clue ("Look at the kettle, Ma!"), and applies a gentle emerald border pulse. Guarantees 100% errorless completion.

Engine 3

30-Day CST Neuroplasticity Accelerator

Quantifies and projects clinical preservation curves against untreated decline trajectories, dynamically expanding the 5-spoke cognitive radar: **Memory (+8%), Recognition (+6%), Reasoning (+7%), Attention (+8%), and Speed (+6%).**

Engine 4

Motivation & Dignity Streak Guard

Forgives missed days caused by rural blackouts, road closures, or illness without penalizing streaks. Generates printable "ASHA Sub-Centre Cognitive Champion" achievement certificates to build self-worth.

8 Technical Implementation Contracts

Supabase Relational Database Schema Additions (schema_v2.sql)

```
CREATE TABLE teamwork_sessions (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  patient_id UUID REFERENCES patients(id) ON DELETE CASCADE,  
  caregiver_id UUID REFERENCES auth.users(id),  
  therapist_id UUID REFERENCES auth.users(id),  
  session_code VARCHAR(8) UNIQUE NOT NULL,      -- e.g. "YS-7821" for pairing  
  mode TEXT DEFAULT 'staging',                  -- 'staging', 'co_op_duo', 'tele_shadow'  
  status TEXT DEFAULT 'pending_review',         -- 'pending_review', 'approved', 'active'  
  target_spoke TEXT DEFAULT 'memory',           -- 'memory', 'attention', 'recognition', etc.  
  option_count INT DEFAULT 4,                   -- 2 for aphasia/severe, 4 for standard  
  bereavement_cleared BOOLEAN DEFAULT FALSE,  
  dialect_checked BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);  
  
CREATE TABLE boost_events (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  patient_id UUID REFERENCES patients(id) ON DELETE CASCADE,  
  session_id UUID REFERENCES game_sessions(id),  
  boost_type TEXT NOT NULL,                     -- 'circadian_micro', 'in_game_scaffold', 'cst_30day_sim'  
  trigger_reason TEXT NOT NULL,                 -- 'latency_spike_2.6s', 'morning_priming', 'manual'  
  pre_latency_ms INT NOT NULL,  
  radar_delta JSONB DEFAULT '{"memory": 8, "recognition": 6, "reasoning": 7, "attention": 8, "speed": 6}':JSONB,  
  recorded_at TIMESTAMPTZ DEFAULT NOW()  
);
```

9 Clinical Persona Matrix

Persona	Clinical Condition & Ground Reality	Core Platform Solution	/teamwork-preview	/boost
Dadi Kamala 72y, Rural Tezpur	Early Alzheimer's, Assamese speaker, low vision, late afternoon sundowning, panic around buzzers.	Bihu Japi & Rhino motifs, 72px tactile buttons, Assamese spoken audio, zero failure buzzers.	Plays in Co-Op Duo mode with grandson Amit reading story prompts.	Circadian 3-min boost at 10:15 AM; in-game lifeline dims wrong choices if latency >2.5s.
Biren Barua 64y, Jorhat	Post-ischemic MCA stroke, right hemiparesis, severe anomic aphasia, high cognitive fatigue.	Multimodal tri-channel naming (photo + slow 0.88x speech audio + semantic options).	SLP in Guwahati previews & calibrates difficulty; caregiver clears photos.	Pre-therapy warmup primes semantic pathways 10 mins before speech sessions.
Priya Phukon ASHA Worker, Majuli	Covers 1,200 rural villagers, zero specialists within 180km, sub-₹8,000 phone.	IsolationForest anomaly alerts (Flag -1), 100% offline IndexedDB queueing, ICMR exports.	Auditions local dialect audio; co-signs verified clinical summaries for MOs.	Initiates 30-day CST Boost protocol for newly screened elders.
Ananya 44y, Daughter, Guwahati	Working daughter, sandwich generation, exhausted by repetitive questions, fear of decline.	5-spoke radar telemetry, AI weekly progress letters in Assamese/English.	Vets sensitive family photos in staging sandbox before mom sees them.	Tracks objective +8% memory lift, verifying therapy efficacy.